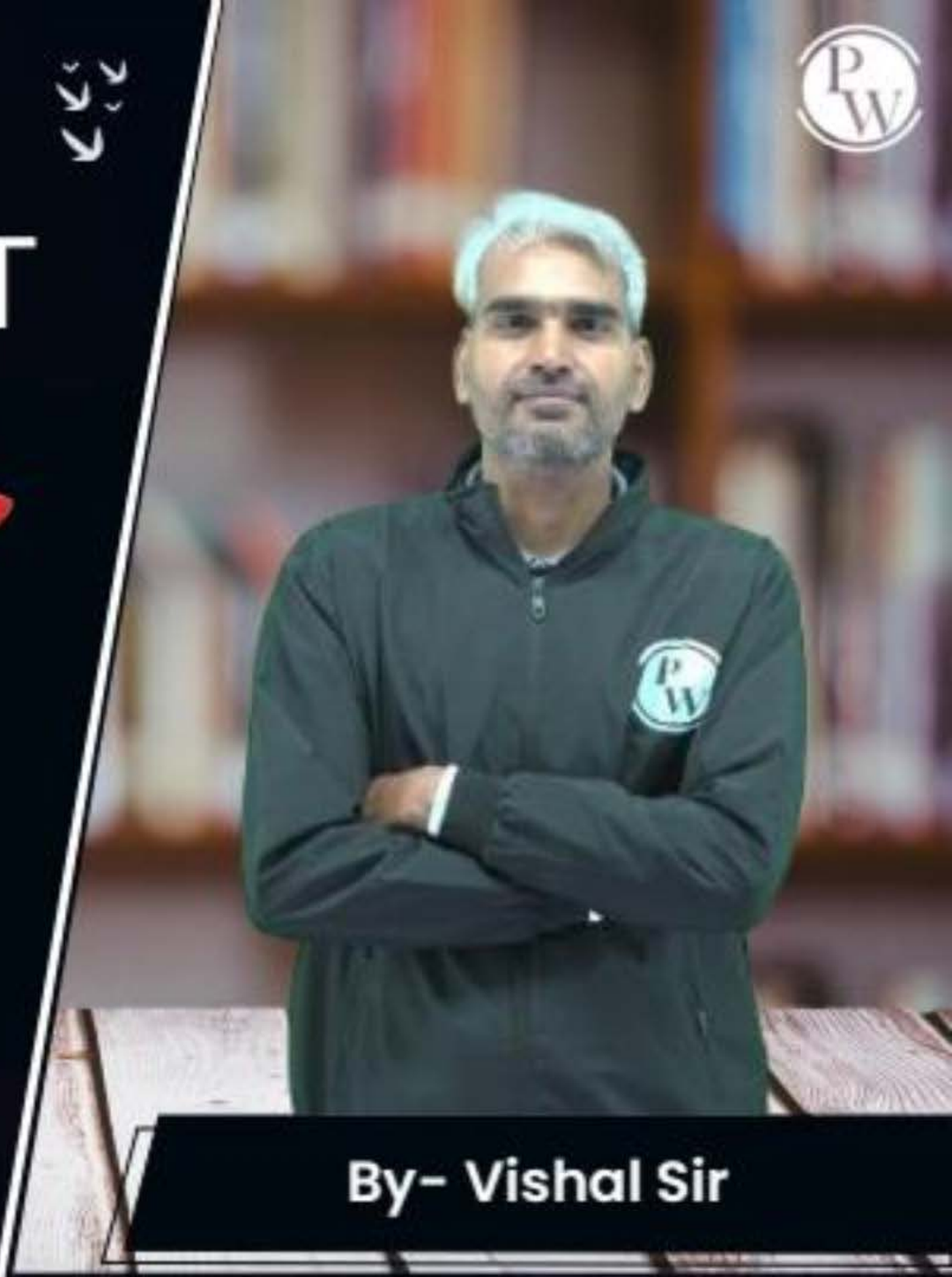
# Computer Science & IT



## Discrete Mathematics

## Combinatorics

Lecture No. 04



By- Vishal Sir

# Recap of Previous Lecture



**Topic**

Solution of recurrence relation using method of characteristic roots

Slide

# Topics to be Covered



**Topic** Solution of recurrence relation using method of characteristic roots

**Topic** Method of undetermined coefficient for particular solution

Slide



# Topic : Method of characteristic roots

Shift operator :

$$a_{n+k} = E^k \cdot a_n$$

$$a_{n+(k-1)} = E^{k-1} \cdot a_n$$

$$\vdots$$

$$\vdots$$

$$a_{n+1} = E^1 \cdot a_n$$

$$a_n = E^0 \cdot a_n = a_n$$



# Topic : Method of characteristic roots

Consider the following linear recurrence relation.

$$\lambda_0 \cdot a_n + \lambda_1 \cdot a_{n-1} + \lambda_2 \cdot a_{n-2} + \cdots + \lambda_k \cdot a_{n-k} = f(n) \quad \text{— eq}^n \text{ (1)}$$

Put $n-k = n$,

i.e., $n = n+k$,

∴ Equation becomes,

$$\lambda_0 \cdot a_{n+k} + \lambda_1 \cdot a_{n+k-1} + \lambda_2 \cdot a_{n+k-2} + \cdots + \lambda_k a_n = F(n) \quad \text{— eq}^n \text{ (2)}$$

It is the new function obtain after replacing n by (n+k) in f(n).



# Topic : Method of characteristic roots

Use shift operator in eqⁿ ② i.e. $a_{n+i} = E^i \cdot a_n$

$$\lambda_0 \cdot (E^k a_n) + \lambda_1 \cdot (E^{k-1} a_n) + \lambda_2 (E^{k-2} a_n) + \cdots + \lambda_k (E^0 \cdot a_n) = F(n)$$

$$E^k \cdot \lambda_0 a_n + E^{k-1} \cdot \lambda_1 a_n + E^{k-2} \cdot \lambda_2 a_n + \cdots + \lambda_k a_n = F(n)$$

$$(E^k \lambda_0 + E^{k-1} \lambda_1 + E^{k-2} \lambda_2 + \cdots + E^0 \lambda_k)\, a_n = F(n)$$

it is a polynomial of degree 'k', let's call it $\phi(E)$

i.e., $\boxed{\phi(E) \cdot a_n = F(n)}$ — eqⁿ ③



# Topic : Method of characteristic roots

* **Characteristic Equation :-**

$\phi(E)$ is a Polynomial of degree = K

$\phi(t) = 0$ is called Characteristic Equation

Roots of Characteristic Equation are called Characteristic roots.

Let the characteristic roots are.

$t_1, t_2, t_3, \ldots, t_K$

'K' characteristic roots, w.r.t. Polynomial of degree = K



# Topic : Method of characteristic roots

## Complementary Function (C.F.)

Complementary function is the solution w.r.t. homogeneous part of linear recurrence relation.
{i.e. w.r.t. $f(n) = 0$}

## Particular Solution :-

Particular solution is the solution w.r.t non-homogeneous part of linear recurrence relation
{i.e. w.r.t. $f(n) \neq 0$}

∴ Complete Solution of linear recurrence Relation = Complementary function + Particular Solution

**Note:-** If given recurrence relation is a homogeneous linear recurrence relation {i.e. $f(n)=0$}, then Complementary function alone gives the Complete solution of the homogeneous linear recurrence relation

$a_n = a_n^H + a_n^P$

$3a_n + 5a_{n-1} + a_{n-2} = 2^n$

$= 0$

$a_n^H \to 0$

$a_n^P \to 2^n$

$3\cdot(a_n^H + a_n^P) + 5(a_{n-1}^H + a_{n-1}^P) + (a_{n-2}^H + a_{n-2}^P)$

$\underbrace{(3\cdot a_n^H + 5a_{n-1}^H + a_{n-2}^H)}_{0} + \underbrace{(3\cdot a_n^P + 5a_{n-1}^P + a_{n-2}^P)}_{2^n} = 2^n$



# Topic : Method of characteristic roots

## Rules to write Complementary function :-

| Characteristic roots | Complementary function |
|---|---|
| ① When all roots are real & distinct<br>i.e., $t_1, t_2, t_3, \ldots, t_k$ | $C_1(t_1)^n + C_2(t_2)^n + C_3(t_3)^n + \ldots + C_k(t_k)^n$<br>$C_1, C_2, \ldots C_k$ are unknown Constants |
| ② When all roots are real and two roots are equal,<br>let, $t_1, t_1, t_3, t_4, \ldots, t_k$ | $(C_1 + C_2 n)(t_1)^n + C_3(t_3)^n + \ldots + C_k(t_k)^n$<br>$C_1, C_2, \ldots C_k$ are Constants |
| ③ When roots are real & three roots are equal,<br>let $t_1, t_1, t_1, t_4, \ldots t_k$ | $(C_1 + C_2 n + C_3 n^2)(t_1)^n + C_4(t_4)^n + \ldots + C_k(t_k)^n$<br>$C_1, C_2, \ldots C_k$ are Constants. |

and so on

Note: We can write the Complementary function in the presence of Complex root as well.
{ Not required for GATE }

Note:- In eq$^{n}$ (1) if $f(n) = 0$, then it becomes homogeneous linear recurrence relation, and Complementary function alone will give us the Complete solution of that linear recurrence relation.

**Note:** In eq$^n$ (1) if $f(n) \neq 0$, then Complete solution is defined as,

$$a_n = \text{Complementary function} + \text{Particular solution}$$

Solution w.r.t homogeneous part

Solution w.r.t. non-homogeneous Part

i.e, if $f(n) \neq 0$, then we need to obtain the Particular solution as well.



# Topic : Method of characteristic roots

## Particular Solution :-

From Eqⁿ ③ $\quad \phi(E) \cdot a_n = F(n)$

i.e. $\quad a_n = \dfrac{F(n)}{\phi(E)}$

Particular solution $(a_n^p)$ is the solution of $\left(\dfrac{F(n)}{\phi(E)}\right)$

P.S. = Solution of $\dfrac{F(n)}{\phi(E)}$

if $f(n) = 0$
then $F(n) = 0$
$\therefore \dfrac{F(n)}{\phi(E)} = \dfrac{0}{\phi(E)} = 0$
Hence when $f(n)=0$, then Particular Solution $= 0$



# Topic : Method of characteristic roots

## Rules for writing Particular Solution When $F(n) = \underbrace{(b)^n}$

↑ Exponential Function with some base = b

$$\text{Particular Solution} = \text{Solution of } \left(\frac{F(n)}{\phi(E)}\right) = \text{Solution of } \left(\frac{b^n}{\phi(E)}\right)$$

Case (1) When '$b$' is not a characteristic root.

i.e, when $\phi(b) \neq 0$

then $\text{Solution of } \left(\frac{b^n}{\phi(E)}\right) = \frac{b^n}{\phi(b)}$

$\therefore$ Particular Solution $(a_n^p) = \frac{b^n}{\phi(b)}$



# Topic : Method of characteristic roots

## Rules for writing Particular Solution When $F(n) = \underbrace{(b)^n}$

↑ Exponential Function with some base = b

$$\text{Particular Solution} = \text{Solution of } \left(\frac{F(n)}{\phi(E)}\right) = \text{Solution of } \left(\frac{b^n}{\phi(E)}\right)$$

**Case ② :-** When 'b' is the characteristic root with multiplicity 'm'.

i.e; $\phi(b) = 0$, and $\phi(E) = (E-b)^m \cdot \psi(E)$, where, $\psi(b) \neq 0$

then Particular Solution $(a_n^p)$ $= \text{Solution of } \left(\frac{b^n}{\phi(E)}\right) = \text{Solution of } \left(\frac{b^n}{(E-b)^m \psi(E)}\right) = \frac{1}{\psi(b)} \left\{ {}^nC_m (b)^{n-m} \right\}$

Q: Find the solution of recurrence relation,

$a_n = 3 \cdot a_{n-1}$ Where $\boxed{a_0 = 1}$

Soluⁿ.

$a_n - 3a_{n-1} = 0$ ← homogeneous

Put $n-1 = n$ { step-1 put, $n-k=n$ }

$a_{n+1} - 3a_n = 0$

Use shift operator i.e. $a_{n+i} = E^i a_n$ { step2: Use shift operator }

$E^1 \cdot a_n - 3 \cdot E^0 a_n = 0$

$E \cdot a_n - 3 \cdot a_n = 0$

$\underbrace{(E-3)}_{\phi(E)} \cdot a_n = 0$

$\phi(E) = E - 3$,

∴ Characteristic equation is $t - 3 = 0$ { $\phi(t) = 0$ is Ch. eqⁿ }

$\boxed{t = 3}$

Characteristic root is $t = 3$

∴ Complementary function (C.F.) = $C_1 \cdot (3)^n$ ($C_1$: Constant)

Given recurrence relation is homogeneous linear recurrence relation ∴ Particular Soluⁿ = 0

Hence, Complete Solution

$a_n = C.F. + P.S.$

$\boxed{a_n = C_1 \cdot (3)^n}$

we know $a_0 = 1$

$\therefore a_0 = C_1 \cdot (3)^0$

$1 = C_1 \cdot (3)^0$

$\boxed{C_1 = 1}$

Hence,

$a_n = 1(3)^n$

$\boxed{a_n = 3^n}$

Q: Find solution of recurrence relation.

$a_n = 2 \cdot a_{n-1} - 1$, where $\boxed{a_1 = 2}$

$a_n - 2a_{n-1} = -1$

Put $n-1 = n$

$a_{n+1} - 2a_n = -1$

Shift operator

$E \cdot a_n - 2a_n = -1$

$(E-2) \cdot a_n = -1$

$\phi(E) = E - 2$

Aux eqn is

$t - 2 = 0$

i.e. Ch root is

$t = 2$.

Hence,

Complementary function $(C.F) = C_1(2)^n$

$\boxed{C.F = C_1(2)^n}$

$a_n^H$

From

$(E-2) \cdot a_n = -1$

$(E-2) \cdot a_n = -1 \cdot \{1^n\}$

$a_n = \frac{-1 \cdot (1)^n}{(E-2)}$ base $b=1$

"$b=1$ is not a characteristic root i.e. $\phi(b) \neq 0$.

$\therefore P.S.(a_n^P) = \text{Soln of } \left(\frac{-1 \cdot (1)^n}{(E-2)}\right)$

$= -1 \cdot \text{Soln of } \left\{\frac{(1)^n}{(E-2)}\right\}$ $\phi(E)$

$= -1 \left\{\frac{(1)^n}{\phi(1)}\right\} = -1 \left\{\frac{(1)^n}{(1-2)}\right\} = 1$

Complete Solution,

$a_n = a_n^H + a_n^P$

$= C.F + P.S.$

$\boxed{a_n = C_1 \cdot (2)^n + 1}$

$a_1 = 2$

$\therefore a_1 = 2 = C_1 \cdot (2)^1 + 1$

$\boxed{C_1 = \frac{1}{2}}$

$\therefore a_n = \frac{1}{2} \cdot (2)^n + 1$

$\boxed{a_n = 2^{n-1} + 1}$

Q: Find the Soluⁿ of recurrence relation,

$a_n = 2 \cdot a_{n-1} + 2^n$ ← f(n) $\quad \boxed{a_0 = 1}$

$a_n - 2a_{n-1} = 2^n$

Put $n-1 = n$, $(n+1) = n$ → F(n)

$a_{n+1} - 2a_n = 2^{n+1}$

using shift operator

$E \cdot a_n - 2 \cdot a_n = 2^{n+1}$

$(E-2) \cdot a_n = 2^{n+1}$

$\phi(E) = (E-2)$

$\therefore \boxed{\phi(E) \cdot a_n = 2(2)^n}$ — eqⁿ ①

Ch. eqⁿ is $t-2=0$

$\therefore$ Ch. root is $t=2$

Hence,

Complementary Function (C.F) $= C_1 \cdot (2)^n$

$\boxed{C.F = a_n^H = C_1 \cdot (2)^n}$

Particular Solution from eqⁿ ①

$\phi(E) \cdot a_n = 2(2)^n$

Particular Soluⁿ = Soluⁿ of $\left(\frac{2 \cdot (2)^n}{\phi(E)}\right)$

$= 2 \cdot \left\{ \text{Soluⁿ of } \frac{(2)^n}{\phi(E) = (E-2)} \right\}$

$b = 2$ & $\phi(b) = \phi(2) = 2-2 = 0$

We know '2' is Ch. root with multiplicity = 1

$\phi(E) = E-2 = (E-2)^1 \cdot 1$

$\phi(E) = (E-2)^1 \cdot \Psi(E)$ when $\Psi(E) = 1$

$= 2 \times \left\{ \text{Soluⁿ of } \frac{(2)^n}{(E-2)^1 \cdot \Psi(E)} \right\}$ ($\Psi(E) \to 1$)

$= 2 \times \left[ \frac{1}{\Psi(2)} \cdot \left( {}^{n}C_{j} \cdot (2)^{n-j} \right) \right]$ ($\Psi(2) \to 1$, $j = 0$ to $m$)

$\therefore P.S.(a_n^P) = 2 \cdot \left\{ \frac{1}{1} \cdot (n \cdot (2)^{n-1}) \right\}$

$P.S.(a_n^P) = n \cdot 2^n$

Complete Soluⁿ

$a_n = CF + P.S$

$= C_1 \cdot 2^n + n \cdot 2^n$

$a_0 = 1$

$\therefore a_0 = 1 = C_1 \cdot 2^0 + 0 \cdot 2^0$

$\boxed{C_1 = 1}$

Hence,

$a_n = 1 \cdot 2^n + n \cdot 2^n$

$= (n+1) \cdot 2^n$

Q: Find the solution of recurrence relation,

$$a_n - 7 \cdot a_{n-1} + 12 \cdot a_{n-2} = 0$$

$a_0 = 1$
$a_1 = 2$

$a_{n+2} - 7a_{n+1} + 12a_n = 0$

$E^2 \cdot a_n - 7 \cdot E \cdot a_n + 12a_n = 0$

$(E^2 - 7E + 12) \cdot a_n = 0$

$\phi(E)$

Ch. eqⁿ is $t^2 - 7t + 12 = 0$

& Ch. root are, $t_1 = 3$, $t_2 = 4$

∴ Complementary function

$C.F = C_1 \cdot (3)^n + C_2 \cdot (4)^n$

homogeneous

∴ $P.S. = 0$

Complete Soluⁿ

$a_n = C.F + P.S.$

$a_n = (C_1 \cdot (3)^n + C_2 (4)^n) + 0$

from, $a_0 = 1$ & $a_1 = 2$,

we get,

$C_1 = 2$ & $C_2 = -1$

∴ $a_n = 2 \cdot (3)^n - 1 \cdot (4)^n$

$$a_n = 2(3)^n - 4^n$$

Q: Find the solution of recurrence relation.

$$T(3^k) = 2 \cdot T(3^{k-1}) + 1$$

$$T(1) = 1$$

Put $T(3^k) = a_k$

$\therefore T(3^{k-1}) = a_{k-1}$

i.e., $a_k = 2 \cdot a_{k-1} + 1$

$(E-2) \cdot a_k = 1$

Ch. root is $t = 2$

$C.F. = C_1 \cdot (2)^k$

recurrence rel$^n$ is a function of 'k'

From $(E-2) \cdot a_k = 1$

$$a_k = \frac{(1)^k}{(E-2)}$$

$\therefore P.S. = $ Soln of $\left(\frac{(1)^k}{(E-2)}\right)$

'1' is not a Ch. root.

$$= \frac{(1)^k}{\phi(1)} = \frac{(1)^k}{(1-2)} = -1$$

$P.S. = -1$

Complete Soln

$a_k = C.F. + P.S.$

$= C_1 \cdot 2^k + (-1)$

$$a_k = C_1 \cdot 2^k - 1$$

$a_k = T(3^k)$

$\therefore T(3^k) = C_1 \cdot 2^k - 1$ & $T(1) = 1$

$3^k = 1$ at $k = 0$.

$T(1) = T(3^0) = C_1 \cdot 2^0 - 1$

$1 = C_1 - 1 \Rightarrow C_1 = 2$

$\therefore T(3^k) = 2 \cdot 2^k - 1 \Rightarrow T(3^k) = 2^{k+1} - 1$

Q: Find the soln of recurrence relation

$$a_n - 2a_{n-1} + a_{n-2} = 0$$

$a_0 = 1$

$a_1 = 2$

homogeneous

$E^2 a_n - 2E a_n + a_n = 0$

$\phi(E) = (E^2 - 2E + 1)$

Ch. eqn is $t^2 - 2t + 1 = 0$

$\therefore$ Ch. roots $t = 1, 1$

two equal roots

$\therefore$ C.F $= (c_1 + c_2 n)(1)^n$

$$\boxed{C.F = c_1 + c_2 n}$$

P.S $= 0$

Complete solution

$a_n = CF + PS$

$= (c_1 + c_2 n) + 0$

$$\boxed{a_n = c_1 + c_2 n}$$

$a_0 = 1 = c_1 \Rightarrow \boxed{c_1 = 1}$

$a_1 = 2 = c_1 + c_2 \cdot 1$

$2 = 1 + c_2 \Rightarrow \boxed{c_2 = 1}$

$a_n = 1 + 1 \cdot n$

$$\boxed{a_n = n + 1}$$

Q: Find the solution of the recurrence relation.

$$a_n - 3a_{n-1} + 2a_{n-2} = 2^n$$

$a_0 = 1$, $a_1 = 2$

$E^2 a_n - 3E a_n + 2a_n = 2^{n+2}$ (because $n = n+2$)

$(E^2 - 3E + 2)\, a_n = 4(2)^n$

Ch eqn is $t^2 - 3t + 2 = 0$

$\therefore$ Ch root are, $t_1 = 2$, $t_2 = 1$

Complementary function

$$C.F = C_1 \cdot (2)^n + C_2 \cdot (1)^n$$

P.S : Soln of $\dfrac{4 \cdot 2^n}{(E^2 - 3E + 2)}$

$= 4 \cdot \left\{ \text{Soln of } \dfrac{(2)^n}{(E^2 - 3E + 2)} \right\}$

'2' is a root with multiplicity = 1

$\therefore \phi(E) = (E-2)^1 \cdot \underbrace{(E-1)}_{\psi(E)}$

P.S : $4 * \left\{ \text{Soln of } \dfrac{(2)^n}{(E-2)^1 \cdot \underbrace{(E-1)}_{\psi(E)}} \right\}$

$= 4 * \left[ \dfrac{1}{\psi(2)} \left\{ n C_1 \cdot (2)^{n-1} \right\} \right]$

$= 4 * \left[ \dfrac{1}{(2-1)} * \left\{ n \cdot (2)^{n-1} \right\} \right]$

P.S $= 2 \cdot n \cdot 2^n$

Complete Soln

$a_n = C.F + P.S$

$= C_1 (2)^n + C_2 + 2 \cdot n \cdot 2^n$

We know $a_0 = 1$ & $a_2 = 2$

$\therefore C_1 = -3$ & $C_2 = 4$

Hence, $a_n = -3 \cdot 2^n + n \cdot 2^{n+1} + 4$

H.W: Find the solution of recurrence relation,

$$a_n^2 - 2a_{n-1}^2 = 1 \qquad a_1 = 2$$

it is not a linear recurrence rel$^n$.

but it can be converted into one

by putting $a_n^2 = x_n$

∴ Equation becomes

$$x_n - 2x_{n-1} = 1$$

$$x_n - 2 \cdot x_{n-1} = 1$$

Put $n-1 = n$

i.e. $n = n+1$

$$x_{n+1} - 2 \cdot x_n = 1$$

Use Shift operator

$x_{n+i} = E^i \cdot x_n$

$$E \cdot x_n - 2 \cdot x_n = 1$$

$$\underbrace{(E-2)}_{\phi(E)} \cdot x_n = 1$$

$\phi(E) = E - 2$

$\therefore$ Ch. eqⁿ is $E - 2 = 0$

& Ch. root is '2'

$\therefore$ Complementary function

$C.F = C_1 \cdot (2)^n$

$$(E-2) \cdot x_n = 1$$

$$x_n = \frac{(1)^n}{(E-2)}$$

base = '1'

and '1' is not a Ch. root,

i.e. $\phi(1) \neq 0$

$\therefore$ P.S = Soluⁿ of $\left(\frac{(1)^n}{(E-2)}\right)$

$$= \frac{(1)^n}{(1-2)} = -1$$

Complete Solution $x_n = C.F + P.S.$

$$x_n = C_1 \cdot 2^n + (-1)$$

$$x_n = C_1 \cdot 2^n - 1$$

$x_n = a_n^2$

$\therefore x_n = a_n^2 = C_1 \cdot 2^n - 1$

$a_1 = 2$

$a_1^2 = (2)^2 = C_1 \cdot 2^1 - 1$

$= C_1 = 5/2$

$a_n^2 = (5 \cdot 2^{n-1} - 1)$

$a_n = \pm \sqrt{5 \cdot 2^{n-1} - 1}$

H.W Find the solution of recurrence relation

$$\sqrt{a_n} - \sqrt{a_{n-1}} - 2\sqrt{a_{n-2}} = 0 \qquad \left\{ \begin{array}{l} a_0 = 1 \\ a_1 = 1 \end{array} \right\}$$

Put $\sqrt{a_n} = x_n$

∴ Equation becomes

$$x_n - x_{n-1} - 2x_{n-2} = 0$$

Put $n-2 = n$

& Use shift operator,

$$E^2 \cdot x_n - E \cdot x_n - 2 \cdot x_n = 0$$

$$(E^2 - E - 2) \cdot x_n = 0$$

$\phi(E) = E^2 - E - 2 = 0$

∴ Ch. eqn is

$t^2 - t - 2 = 0$

$t^2 - 2t + t - 2 = 0$

Ch roots are

$t_1 = 2, \; t_2 = -1$

∴ Complementary function

$CF = C_1 \cdot (2)^n + C_2 \cdot (-1)^n$

$f(n) = 0$, ∴ P.S. = 0

Complete Soluⁿ

$x_n = CF + \cancel{PS} \to 0$

$x_n = C_1 \cdot 2^n + C_2 \cdot (-1)^n$

$x_n = \sqrt{a_n}$

$\therefore x_n = \sqrt{a_n} = C_1 (2)^n + C_2 \cdot (-1)^n$

$a_0 = 1, \; \therefore \sqrt{a_0} = C_1 \cdot 2^0 + C_2 (-1)^0$

$1 = C_1 + C_2$

$a_1 = 1, \; \therefore \sqrt{a_1} = C_1 \cdot 2 + C_2 (-1)^1$

$1 = 2C_1 - C_2$

$C_1 = \frac{2}{3} \quad C_2 = \frac{1}{3}$

$\sqrt{a_n} = \frac{2}{3}(2)^n + \frac{1}{3}(-1)^n$

$a_n = \left(\frac{2}{3}(2)^n + \frac{1}{3}(-1)^n\right)^2$

# Divide & Conquer Relation :

A relation of the type

$$T(n) = c \cdot T\left(\frac{n}{d}\right) + f(n)$$

is called divide & conquer relation

Where c & d are constants.

We can convert the given relation into linear recurrence relation by substituting

$$n = d^k$$

i.e Equation becomes

$$T(d^k) = c \cdot T\left(\frac{d^k}{d}\right) + g(k)$$

function after substituting $n = d^k$ in $f(n)$

$$\boxed{T(d^k) = c \cdot T(d^{k-1}) + g(k)}$$

We have already solved such recurrence relation

Q: Find the solution of recurrence relation

$T(n) = 7 \cdot T\left(\frac{n}{3}\right) + 2n$ where $T(1) = 2$

$2n$ = f(n), $2 \cdot 3^k$ = g(k)

Put $n = 3^k$

$T(3^k) = 7 \cdot T(3^{k-1}) + 2 \cdot 3^k$

Put $T(3^k) = x_k$

$x_k = 7 \cdot x_{k-1} + 2 \cdot 3^k$

Converted into linear Recurrence relation →

$x_k - 7 \cdot x_{k-1} = 2 \cdot 3^k$

Put $k-1 = k$

i.e., $k = k+1$

$x_{k+1} - 7 \cdot x_k = 2 \cdot (3)^{k+1}$

Use shift operator

$E \cdot x_k - 7 \cdot x_k = 2 \cdot (3)^{k+1}$

$\underbrace{(E-7)}_{\phi(E)} \cdot x_k = 2(3)^{k+1}$

Ch. eqn $t - 7 = 0$,

Ch. root is $t_1 = 7$,

∴ Complementary function

$C.F = C_1 \cdot (7)^k$

$\phi(E) \cdot x_k = 2 \cdot (3)^{k+1}$

$x_k = \frac{6 \cdot (3)^k}{\phi(E)}$

Particular Solution = Solution of $\left(\frac{6 \cdot (3)^k}{E-7}\right)$

$P.S = 6 * \left\{ \text{Solution of } \frac{(3)^k}{(E-7)} \right\}$

base '3' is not a Ch. root

∴ $P.S = 6 * \left\{ \frac{(3)^k}{\phi(3)} \right\}$

$= 6 * \left\{ \frac{(3)^k}{(3-7)} \right\} = \frac{-3}{2} \cdot (3)^k$

$P.S = \frac{-3}{2}(3)^k$

Complete Solution $x_k = C.F + P.S.$

$$x_k = C_1 \cdot (7)^k + \left(\frac{-3}{2} \cdot (3)^k\right)$$

$x_k = T(3^k)$

$x_k = T(3^k) = C_1 \cdot (7)^k - \frac{3}{2}(3)^k$

$T(1) = 2$, $3^k$ will be '1' at $k = 0$

$$T(1) = C_1 \cdot (7)^0 - \frac{3}{2} \cdot (3)^0$$

$$2 = C_1 - \frac{3}{2}$$

$$\boxed{C_1 = 7/2}$$

$$T(3^k) = \frac{7}{2} \cdot (7)^k - \frac{3}{2} \cdot (3)^k$$

We know $3^k = n$

$\therefore \log_3(3^k) = \log_3(n)$

$$\boxed{k = \log_3 n}$$

$$\therefore T(n) = \frac{7}{2} \cdot (7)^{\log_3 n} - \left(\frac{3}{2} \cdot n\right)$$

$$\boxed{T(n) = \frac{7}{2} \cdot (7)^{\log_3 n} - \frac{3n}{2}}$$

Q: Find the solution of recurrence relation,

$T(n) = 2 \cdot T\left(\frac{n}{2}\right) + (n+1)$ where $T(1) = 0$

Put $n = 2^k$

$T(2^k) = 2 \cdot T(2^{k-1}) + (2^k + 1)$

Put $T(2^k) = x_k$

$x_k = 2 \cdot x_{k-1} + 2^k + 1$

$x_k - 2x_{k-1} = 2^k + 1$

Put $k-1 = k$, i.e. $k = k+1$

$x_{k+1} - 2 \cdot x_k = 2^{k+1} + 1$

$E \cdot x_k - 2 \cdot x_k = 2^{k+1} + 1$

$\underbrace{(E-2)}_{\phi(E)} \cdot x_k = 2^{k+1} + 1$

Ch. eqn is $t - 2 = 0$

Ch. root is $t_1 = 2$.

$C.F = C_1(2)^k$

$$\underbrace{(E-2)}_{\phi(E)}\, x_k = 2^{k+1} + 1$$

$$\text{Particular Solut} = \text{Solun of}\left(\frac{2^{k+1}+1}{(E-2)}\right)$$

$$= \text{Solun of}\left(\frac{2^{k+1}}{E-2} + \frac{1}{(E-2)}\right)$$

$$= \text{Solun of}\left(\frac{2^{k+1}}{E-2}\right) + \text{Solun of}\left(\frac{1}{E-2}\right)$$

$$2 * \text{Solun of}\left(\frac{2^k}{E-2}\right) + \text{Solun of}\left(\frac{(1)^k}{E-2}\right)$$

base '2' is Ch. root with multiplicity = 1

base '1' is not a Ch. root

$$\phi(E) = (E-2)^1 \cdot 1$$

$$\psi(E)$$

$$2 * \left\{\frac{1}{\psi(2)} \cdot {}^{k}C_1 \cdot (2)^{k-1}\right\} + \left(\frac{1}{\phi(1)}\right)$$

$$= 2 * \left(\frac{1}{1} * k * 2^{k-1}\right) + \left(\frac{1}{(1-2)}\right)$$

$$= k \cdot 2^k \qquad +(-1)$$

$$\boxed{P.S = k \cdot 2^k - 1}$$

## Complete Soluⁿ

$$x_k = C.F + P.S.$$

$$x_k = C_1(2)^k + k \cdot 2^k - 1$$

$$x_k = T(2^k) = C_1(2)^k + k \cdot 2^k - 1$$

$2^k = 1$ at $k = 0$

$T(1) = 0$

$$T(1) = 0 = C_1(2)^0 + 0 \cdot 2^0 - 1$$

$$0 = C_1 - 1$$

$$\boxed{C_1 = 1}$$

$$T(2^k) = 2^k + k \cdot 2^k - 1$$

$2^k = n$, $\therefore$ $\boxed{k = \log_2 n}$

Hence,

$$T(n) = n + (\log_2 n) \cdot n - 1$$

$$\boxed{T(n) = n + n \cdot \log_2 n - 1}$$

Q. Let $T(n)$ be the recurrence relation defined as

$$T(n) = 5 \cdot T(n-1) - 6 \cdot T(n-2)$$ Where, $T_0 = 1$, $T_1 = 2$

Which of the following statement is true.

(a) $T(n) = \Theta(n \cdot 2^n)$

(b) $T(n) = \Theta(2^n)$ ✓

(c) $T(n) = \Theta(3^n)$

(d) $T(n) = \Theta(n \cdot 3^n)$

let $T(n) = a_n$

$a_n - 5a_{n-1} + 6a_{n-2} = 0$ ← Homogeneous

Ch. eqn is

$t^2 - 5t + 6 = 0$

Ch. root are

$t_1 = 3, t_2 = 2$

C.F $= C_1(3)^n + C_2(2)^n$

P.S $= 0$

∴ Complete Soln

$a_n = C_1 \cdot (3)^n + C_2(2)^n$

$a_n = T(n) = C_1 \cdot (3)^n + C_2 \cdot (2)^n$

Put $T_0 = 1$ & $T_1 = 2$

we get $C_1 = 0$ & $C_2 = 1$

∴ $T(n) = 1 \cdot (2)^n$



# Topic : Method of undetermined coefficient

Method of undetermined coefficient can be used to identify the solution of non-homogeneous part of linear recurrence relation when function f(n) is neither a constant nor an exponential function

Constant = $\mathcal{C}$

$f(n) = C \cdot (1)^n$

Exponential function = $C \cdot (b)^n$ (Constant)

$f(n) = C \cdot (b)^n$



# Topic : Method of undetermined coefficient

- Let, $\lambda_0 a_n + \lambda_1 a_{n-1} + \lambda_2 a_{n-2} + \cdots + \lambda_k a_{n-k} = f(n)$ be the linear recurrence relation.

- and after replacing n by 'n+k'. and after using shift operator eqⁿ becomes,

$$\phi(E)\, a_n = F(n)$$

function after substituting 'n' by 'n+k'

- We also know $\phi(t) = 0$ is the ch eqⁿ, and ch. root are roots of the eqⁿ $\phi(t) = 0$.

# Topic : Method of undetermined coefficient

## Rules for writing the particular Soln :

| $f(n)$ : function in original linear recurrence relⁿ | $\phi(E)$ | Particular Solⁿ. |
|---|---|---|
| Exponential + Polynomial of deg = s<br>$(b)^n (C_0 n^s + C_1 n^{s-1} + \dots + C_s)$<br>$C_0, C_1, \dots$ etc are constants | if $\phi(b) = 0$,<br>i.e. 'b' is a characteristic root (let with multiplicity = m) | $(b)^n (A_0 n^s + A_1 n^{s-1} + \dots + A_s)(n)^m$<br>$A_0, A_1, A_2, \dots$ etc are unknown coefficient |
| | if $\phi(b) \neq 0$<br>i.e., 'b' is not a Characteristic root | $(b)^n (A_0 n^s + A_1 n^{s-1} + \dots A_s)$ ~~n~~ '1'<br>$A_0, A_1, \dots A_s$ are unkown |

# Topic : Method of undetermined coefficient

## Rules for writing the particular Soln :

| $f(n)$ : function in original linear recurrence rel$^n$ | $\phi(E)$ | Particular Soln. |
|---|---|---|
| $(C_0 n^s + C_1 n^{s-1} + \dots + C_s)$ (Polynomial of deg $= s$) <br> $C_0, C_1, \dots$ etc are constants <br> If nothing is there, then we can assume that $(1)^n$ is there | if $\phi(1) = 0$, <br> i.e. '1' is a characteristic root (let with multiplicity $= m$) | $\cdot (A_0 n^s + A_1 n^{s-1} + \dots + A_s)(n)^m$ <br> $A_0, A_1, A_2, \dots$ etc are unknown coefficient |
| | if $\phi(1) \neq 0$ <br> i.e., '1' is not a Characteristic root | $(A_0 n^s + A_1 n^{s-1} + \dots A_s)\, n^{\text{'1'}}$ <br> $A_0, A_1, \dots A_s$ are unkown |

#Q. Find the solution of recurrence relation

$a_n - 3a_{n-1} = n+3$

$a_{n+1} - 3a_n = (n+1)+3$

$E.a_n - 3.a_n = n+4$

$(E-3).a_n = n+4$

$\phi(E)$

Ch. eqⁿ is $t-3=0$

Ch. root is $t=3$

$\therefore$ C.F. $= C_1.(3)^n$

Particular Solⁿ:

$f(n) = (n+3)$

Polynomial of deg = 1

No. Exponential function $\therefore (1)^n$ is there

Check $\phi(1)$

$\phi(1) = 1-3 = -2 \neq 0$

$\therefore$ Particular Solⁿ $a_n^{(P)} = An+B$ — Polynomial of deg: 1

Unknown

Particular Soln $a_n^{(P)}$ satisfies the given recurrence relation

i.e; $a_n^{(P)} - 3 \cdot a_{n-1}^{(P)} = n+3$

$a_n^{(P)} = An+B$

$\therefore a_{n-1}^{(P)} = A(n-1)+B$

$\therefore (An+B) - 3(A(n-1)+B) = n+3$

Put $n=0$,

$B - 3(-A+B) = 3$

$3A - 2B = 3$ — ①

Put $n=1$,

$(A+B) - 3(A \cdot 0 + B) = 4$

$A - 2B = 4$ — ②

By eqn ① & ②

$A = \frac{-1}{2}$ $\quad B = \frac{-9}{4}$

$\therefore P.S = An+B$

$= \frac{-1}{2} \cdot n + \left(\frac{-9}{4}\right)$

$P.S = \left(\frac{-n}{2} - \frac{9}{4}\right)$

Complete Soln

$a_n = CF + PS$

$a_n^{(H)} + a_n^{(P)}$

$$a_n = C_1 \cdot (3)^n + \left(\frac{-n}{2} - \frac{9}{4}\right)$$

$C_1$ Can be Calculated using initial Condn

#Q. Find the solution of recurrence relation

$a_n - a_{n-1} = (n+2)(1)^n$

Ch root $t = 1$

$\therefore CF = C_1 \cdot (1)^n$

$a_n^{(H)} = C_1$

'1' is root with multiplicity = 1

$\therefore P.S = (An+B) \cdot (n)^1$

$a_n^{(P)} = (An+B) \cdot n$

it will satisfy given Recurrence relation

i.e. $a_n^{(P)} - a_{n-1}^{(P)} = n+2$

$n(An+B) - (n-1) \cdot (A(n-1)+B) = n+2$

H.W. Calculate A & B

#Q. Find the solution of recurrence relation

$$a_n - 2.a_{n-1} + a_{n-2} = 3n+5$$

$(3n+5)\cdot(1)^n$

Ch. eqn $t^2 - 2t + 1 = 0$,

$\therefore$ Ch roots

$t_1 = 1, \quad t_2 = 1$

two equal roots

'1' is a root with multiplicity = 2

$\therefore$ P.S $= (An+B)\cdot n^2$

$C.F = (C_1 + C_2 n)\cdot(1)^n$

$a_n^{(H)} = C_1 + n.C_2$

$a_n^{(P)} = (An+B)\cdot n^2$

$(An+B)\cdot n^2 - 2.\{(A(n-1)+B)\cdot n^2\} + ((A(n-2)+B)\cdot n^2) = 3n+5$

Hw. Calculate A & B

#Q. Find the solution of recurrence relation

$a_n - a_{n-1} = 2^n.n$

$= 2^n.(1.n+0)$ ← Polynomial of deg = 1

2 is not a root

Ch root $t = 1$

$C.F = C_1.(1)^n$

$\therefore P.S = (2)^n.(An+B).(n^0)$ → 1

$a_n^{(H)} = C_1$

$a_n^{(P)} = 2^n.(An+B)$

#Q. Find the solution of recurrence relation

$$a_n - 3.a_{n-1} = 3^n(n+2)$$

Ch. root $t=3$

$\therefore C.F = C_1.(3)^n$

3 is a root with multiplicity $= 1$

$\therefore P.S = (3)^n.(An+B).n^1$

$$\boxed{a_n^{(H)} = C_1.(3)^n}$$

$$a_n^{(P)} = n.3^n.(An+B)$$



# 2 mins Summary

**Topic** Solution of recurrence relation using method of characteristic roots

**Topic** Method of undetermined coefficient for particular solution

Slide

# THANK - YOU